PHP Introduction (Detailed Guide for PDF)

What is PHP?

PHP (Hypertext Preprocessor) is a server-side scripting language used to create dynamic web pages and web applications.

History of PHP

- Created by Rasmus Lerdorf in 1994
- Originally stood for Personal Home Page
- Widely used in web development

Why Use PHP?

- Easy to learn
- Open source
- Works well with databases (MySQL)
- Runs on servers

Features of PHP

- Server-side scripting
- Platform independent
- Supports databases
- Fast execution

Basic PHP Syntax

```
<?php
 echo "Hello, World!";
?>
```

Variables

```
$name = "John";
$age = 25;
```

Data Types

- String
- Integer
- Float
- Boolean
- Array
- Object

Operators

- Arithmetic (+, -, *, /)
- Comparison (==, !=)
- Logical (&&, ||)

Conditional Statements

```
if ($age > 18) {
  echo "Adult";
} else {
  echo "Minor";
}
```

Loops

For Loop

```
for ($i = 0; $i < 5; $i++) {
  echo $i;
}
```

While Loop

```
$i = 0;
while ($i < 5) {
  $i++;
}
```

Functions

```
function greet() {
  echo "Hello";
}
```

Arrays

```
$numbers = array(1, 2, 3);
```

Working with Forms

```
$name = $_POST['name'];
```

Connecting to Database

```
$conn = mysqli_connect("localhost", "root", "", "test");
```

Example Project

```
<?php
 echo "Welcome to PHP";
?>
```

Advantages of PHP

- Easy to use
- Free and open source
- Strong database support

Disadvantages of PHP

- Security issues if not coded properly
- Less modern compared to newer languages

Best Practices

- Validate user input
- Use secure coding techniques
- Keep code organized